

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	WTID-2026-2950
Project Title	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform debt relief and compromise management using generative AI and quantitative analytics, boosting negotiation efficiency and borrower financial rehabilitation. It tackles modern debt settlement inefficiencies, promising better distress evaluation, legally secure hardships letters automation, and reduced litigation risks.

Project Overview	
Objective	The primary objective is to revolutionize the lender-borrower resolution space by implementing specialized financial stress engines and generative AI workflows, ensuring customized and fast settlement recommendations.
Scope	The project comprehensively assesses and enhances the automated loan compromise channel, incorporating asynchronous web pipelines for a robust and secure legal relief system.
Problem Statement	
Description	Addressing the total absence of objective, borrower-centric calculation frameworks and formal legal negotiation tools in traditional platforms, which leaves distressed users vulnerable to aggressive collection loops .
Impact	Solving these critical issues will result in improved user psychological relief, transparent debt recovery pathways, and an overall reduction in retail loan default gridlocks.
Proposed Solution	
Approach	Employing mathematical rule engines to parse dynamic Debt-to-Income weights alongside Google Gemini SDK orchestration to auto-synthesize customized bank-ready hardship declaration models.
Key Features	- Implementation of an automated debt-stress calculation tiering dashboard.

	- Real-time Gemini AI text synthesis for professional hardship resolution letter logs. - Persistent SQLite historical log caching to safely isolate and track calculation variations over time.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Web Server Hosting / Local machine setup for backend compilation	Dual-Core CPU / Standard Local Machine
Memory	System operational volatile memory workspace	8 GB
Storage	Cloud or local block disk space for tracking local history ledgers	256 GB SSD
Software		
Frameworks	Core API System Router & UI Application structure	FastAPI (Python 3.11+), React.js
Libraries	Third-party client toolkits and AI orchestration wrappers	google-genai SDK, SQLAlchemy ORM, Pydantic, Axios
Development Environment	Code Editor and development runtime interface setup	Visual Studio Code, Vite Build Tool
Data		
Data	Input storage format, system query schema parameters	Local SQLite relational file, JSON structures, Pydantic data models